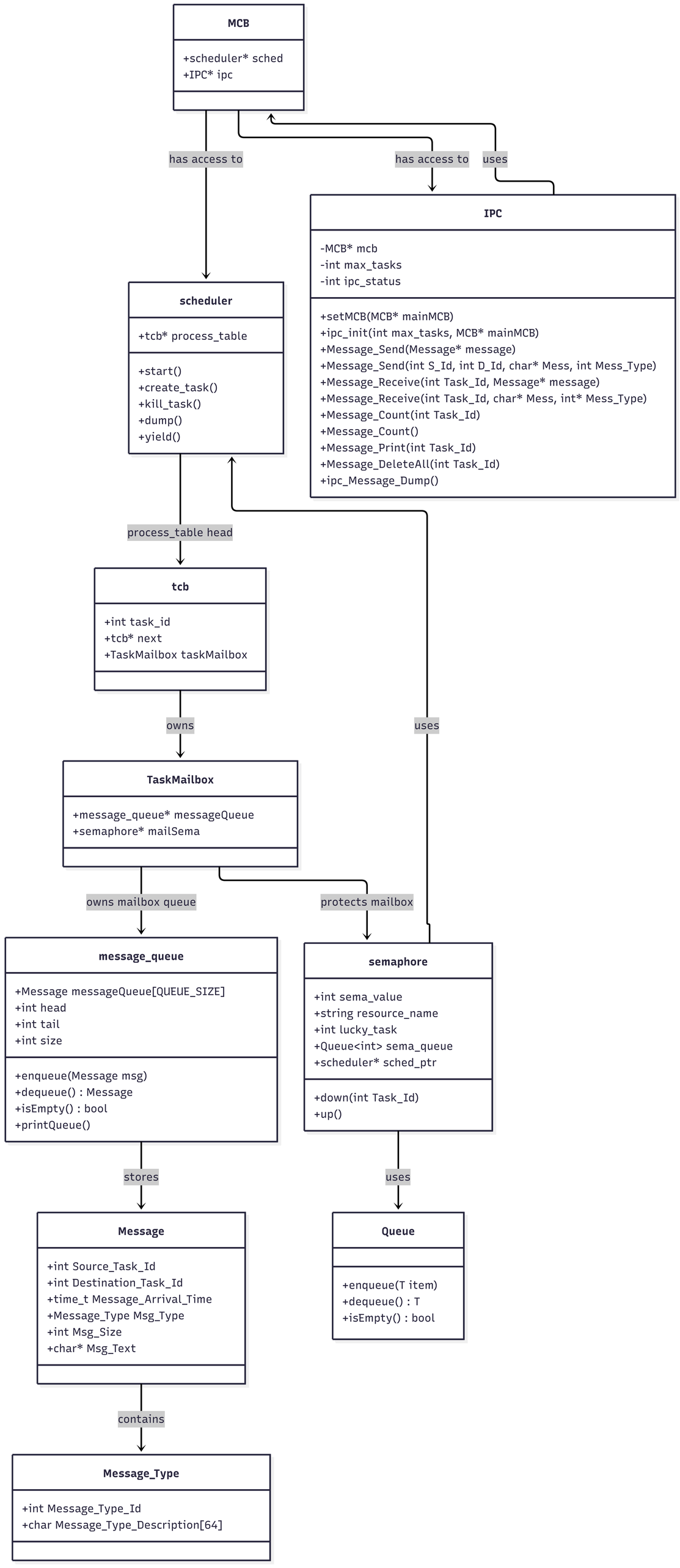
MCB
+scheduler* sched
+IPC* ipc
has access to
has access to
uses
IPC
-MCB* mcb
-int max_tasks
-int ipc_status
+setMCB(MCB* mainMCB)
+ipc_init(int max_tasks, MCB* mainMCB)
+Message_Send(Message* message)
+Message_Send(int S_Id, int D_Id, char* Mess, int Mess_Type)
+Message_Receive(int Task_Id, Message* message)
+Message_Receive(int Task_Id, char* Mess, int* Mess_Type)
+Message_Count(int Task_Id)
+Message_Count()
+Message_Print(int Task_Id)
+Message_DeleteAll(int Task_Id)
+ipc_Message_Dump()
scheduler
+tcb* process_table
+start()
+create_task()
+kill_task()
+dump()
+yield()
process_table head
tcb
+int task_id
+tcb* next
+TaskMailbox taskMailbox
owns
uses
TaskMailbox
+message_queue* messageQueue
+semaphore* mailSema
owns mailbox queue
protects mailbox
message_queue
+Message messageQueue[QUEUE_SIZE]
+int head
+int tail
+int size
+enqueue(Message msg)
+dequeue() : Message
+isEmpty() : bool
+printQueue()
semaphore
+int sema_value
+string resource_name
+int lucky_task
+Queue<int> sema_queue
+scheduler* sched_ptr
+down(int Task_Id)
+up()
stores
uses
Message
+int Source_Task_Id
+int Destination_Task_Id
+time_t Message_Arrival_Time
+Message_Type Msg_Type
+int Msg_Size
+char* Msg_Text
Queue
+enqueue(T item)
+dequeue() : T
+isEmpty() : bool
contains
Message_Type
+int Message_Type_Id
+char Message_Type_Description[64]